Note: This document was presented to Dave Clark and outlines the unique value proposition of Vendor Flex.

Doc Ninja Thoughts:
Document Strengths: The author articulates what the document will be about, introducing the topics that will be discussed and providing relevant background information. The doc is well-organized to meet the purpose of reviewing a solution's progress, with plain global English.
Areas to Develop: The purpose of the doc focuses on the perspective of what the author wants to cover in the doc itself. What the author wants the audience to do with this information is not stated clearly right away, so the author can have a clearer call to action. The author can improve conciseness by keeping sentences under 25 words and using active voice. For example, the last sentence of the first paragraph totals 140 words. Each item in the list can become its own sentence.Some jargon is explained, but not all (e.g. "IB").

Section 1: Executive Summary

Vendor Flex (VF) is an adaptable and just-in-time supply chain solution that lowers fulfillment costs for Amazon by directly shipping customer orders out of a vendor's warehouse. As the Amazon Fulfillment Network (AFN) emphasizes speed and selection in support of one-day-core (1DC), Sortable and Non-sort network capacity requirements continue to increase. VF is uniquely positioned to compliment AFN in 2019 by shipping 25.3M units across 66 nodes, generating a total savings of $87.5MM at a FulCPU savings of $3.46. VF will drive down fulfillment costs for a subset of selection that is available in both VF and AFN by: i) providing 7.28M net cube alleviation (Sortable 1.28M and Non-sort 6.02M) through rent free space across 66 nodes, enabling AFN to expand selection for 1DC; ii) supporting 1DC with a two-hour (2HR) Fast Track (FT) SLA with a 1DC penetration of 13.5% (August 2019); iii) providing rapid replenishment capabilities and mitigating out-of-stock (OOS) risks with a vendor lead time (VLT) of 2.98 days and 18.3 turns vs. 10.21 and 5.0 respectively for AFN (same ASINs); and iv) providing a low cost fulfillment solution for Subscribe and Save (SnS) selection (53% of all VF units ship SnS), which has less speed affinity but drives complexity (SnS orders are encumbered in inventory up to nine days).

In 2020, VF will ship 40.0M units, across 91 nodes and alleviate 10.69M net cube (Sortable 1.88M and Non-sort 8.81M) during Peak, while being highly selective on vendor partnerships.

This document provides an in-depth review of VF by describing how it compliments and enhances AFN capabilities, and discusses VF alignment to network-wide initiatives such as 1DC and Amazon Internal Carriers (IC) expansion.

Section 2: Vendor Flex Financials

Overview: August YTD, VF has a FulCPU of $5.14 compared to $8.60 for the same ASINs in AFN, with August YTD FulCPU savings of $3.46 (+42% YoY) compared to AFN. FulCPU savings is comprised of savings across inbound transportation (IB TransCPU), fixed (FCPU) and variable (VCPU). Table 1 highlights August YTD FulCPU savings compared to AFN. The savings methodology outlined below is in alignment with SCOT, Ops and Retail Finance.

Table 1 – August YTD VF vs. AFN Savings

FulCPU Bucket	VF	AFN	Savings
IB Trans CPU	$ -	$ 0.35	$ 0.35
Fixed	$ 0.14	$ 3.17	$ 3.03
Variable	$ 0.54	$ 1.18	$ 0.65
OB TransCPU	$ 4.42	$ 3.90	$ (0.52)
UPB	$ 0.04	$ -	$ (0.04)
Total	**$ 5.14**	**$ 8.60**	**$ 3.46**

Inbound Freight Savings: IB TransCPU represent 0% of VF total FulCPU, accounting for +10.1% of VF FulCPU savings with an average savings of $0.35/u. VF nodes are modeled with a zero IB cost (located within a vendor's warehouse) and we compare ourselves to the AFN by calculating ASIN-level IB freight costs by vendor.

Fixed Avoidance: FCPU avoidance is the leading financial benefit for VF. Fixed represents 2.7% of VF total FulCPU, accounting for +87.5% of VF FulCPU savings with an average savings of $3.03/u. To evaluate VF fixed cost avoidance, we compare how much inventory cube AFN would require for Peak based on VF Peak shipped units and AFN inventory weeks cover (same ASINs). This is broken down at the ASIN/Node level, while factoring Sortable and Non-sort percentage.

Variable Cost Savings: VCPU (accounts for $29.50 all-in wage rate) and represents 10.5% of VF total FulCPU accounting for +18.8% of VF FulCPU savings with an average savings of $0.65/u. Average Non-sort savings compared to AFN is $0.95 and $0.53 for Sortable. Non-sort savings are higher, as a result of: i) high velocity ship-in-own-container (SIOC) selection (45.0% of overall VF units and 85.0% of Non-sort are SIOC) that is cross-docked in pallet quantities; and ii) bulk pack functionality, which allows us to pack out single ASIN pallets in one-step. Additional levers we utilize to keep variable costs low, include i) temporary labor to ramp up and down headcount depending on daily charge (25% off Peak for VF vs. 15% for AFN); ii) labor sharing between VF nodes that are clustered together; iii) vendor funded labor, which has been rolled out across five Private Brands (PB) nodes, with the vendor covering all VF labor costs (driving incremental variable cost savings of $0.40). We will look to deploy this model on future VF launches; and iv) simplified process paths within VF nodes due to smaller footprint, where one associate handles multiple process paths and can quickly flex between them as required.

Outbound TransCPU: OB TransCPU represents 86.0% of total VF FulCPU, accounting for a -15.2% impact to total savings with an average of -$0.52/u. To evaluate OB TransCPU to AFN[1], total ship costs for the same ASINs shipped across AFN are compared during the same time period, without any mix adjustment.

Units per Box: UPB represents 0.8% of total FulCPU, accounting for a -1.2% impact to total savings with an average impact of -$0.04/u. Our UPB impact (1.28 VF vs. 1.41 AFN) is evaluated at the ASIN level by comparing the UPB of the same ASINs within AFN.

[1] $0.52 OB TransCPU delta between VF and AFN, is based on actual ship costs for the same ASINs. In a scenario where VF did not exist, F2P simulation shows an increase in OB delta (see FAQ 4 for details).

Section 3: Value Proposition

Overview: Over the last three years, we have experimented across product types and businesses to define our value proposition and focus on the product mix best suited for VF operations. Based on learnings, vendor and ASIN selection criteria have been developed to right size our fulfillment mix (Appendix A). Key themes that reinforce the value of the VF program are outlined below.

Cube Alleviation – Leveraging VF to Reduce Peak Capacity Needs: Incremental storage demand is growing faster than throughput demand by 6% annually. The growth differential is expected to accelerate as 1DC expands. As a result, the 2020 to 2022 P50 build plan calls for an incremental 45 buildings (25 Sortable and 20 Non-sort). VF is positioned to alleviate cube constraints in AFN, primarily in Non-sort. VF provides 770K total ft^2 of rent-free space, across 66 nodes with a total of 1.84M net cube (84% of total space dedicated to Non-sort at 1.55M net cube). If VF did not exist, while accounting for AFN turns (Table 2), AFN would require an additional 7.28M net cube during Peak week (1.28M Sortable and 6.02M Non-sort). To ensure that VF-fulfilled demand and resulting AFN cube is accounted for, VF demand forecasts are provided to SCOT IPB and Capacity Planning on a monthly basis. We continue to engage SCOT teams, specifically with Buying and Placement, to incorporate the financial benefits of VF within the MEPv2 model and refine the VF legacy placement algorithm to further maximize VF turns. We believe that VF, with benefits such as reduced VLT and fixed cost avoidance, is an enabler to reducing Non-sort cube needs across AFN.

Table 2 – VF vs. AFN Cube and Turns Comparison

		ACU		Inv Cube		Inv Units		Inv Turns		Peak OB	Inv Cube Avoidance
Sort Type	ASINs	VF	AFN	VF	AFN	VF	AFN	VF	AFN	VF	AFN
Non-Sort	808	1.56	1.64	649,605	5,404,337	417,331	3,294,738	25.4	5.3	393,775	6,002,446
Sort	3890	0.18	0.12	200,415	2,123,642	1,112,949	17,747,311	16.3	4.8	657,976	1,281,013
Total	4698	0.55	0.36	850,020	7,527,979	1,530,280	21,042,049	18.3	5.0	1,051,751	7,283,459

*Inventory Snapshot of 9/05 with trailing four weeks of shipped units data

Speed – Enable 1DC Across VF: To support network-wide expansion of 1DC, VF nodes were enabled for 1DC shipping in two phases. During phase one (week 14), all VF nodes were enabled with a 2HR FT SLA, while retaining 5-day operations. With phase two (starting week 18), 43 VF nodes (65% of existing nodes) were transitioned to a 7-day operation and by week 44, 61 nodes (92%) will run a 7-day operation. Factors contributing to five nodes not enabled for 7-day operations include: i) three nodes having extremely low volume not justifying additional labor cost (i.e., projected 120 average units shipped on weekends per node); ii) one node constrained due to vendor not operating on weekends; and iii) one node facing vendor profitability issues across Amazon. To optimize 1DC entitlement, CPTs have been updated to maximize 1DC penetration, sort center (SC) connectivity has increased, and incremental nodes with existing vendors have been launched. August 1DC penetration is 13.5% of all units within VF (+10% vs. week 13), with average OB TransCPU of $3.85. Conversely, 1DC penetration for the same ASINs in AFN is 37% of all units with an average OB TransCPU of $3.41. The 1DC delta between VF and AFN is a function of: i) limited selection within a node (i.e., only have capabilities to fulfill a single vendor's selection); ii) limited topology – for greater than ten units in inventory, average FC per FNKSU for Non-sort is 4 for VF vs. 35 in AFN, Sortable is 2 for VF vs. 43 in AFN; and iii) VF inventory for ASINs that have the highest 1DC penetration in AFN have 95% of inventory encumbered to SnS orders. To increase our 1DC capabilities, we will i) launch incremental nodes with vendors who have the ability to open at least 2 nodes; ii) evaluate launching second shift operation to maximize 1DC charge; and iii) enabling all VF nodes with connections to local SCs. While 1DC with 7-day operations will be a prerequisite for all new VF launches, limitations exist, specifically: i) vendors that do not carry speed sensitive selection, as in the case of nodes with First Quality (FQ) who have high SnS penetration (98% of all units); and ii) instances where vendors do not operate on weekends and are not willing to change their operating schedule. In these instances, the benefit of launching a 5-day operation vs. not launching at all will be evaluated.

Rapid Replenishment – Leverage VF's Low VLT: VF has an average VLT of 2.98[2] days (August YTD average) vs. 10.21 for the same ASINs in AFN. The reduced VLT provides an opportunity to buy over shorter planning horizons, protect against OOS situations in AFN, and support new product launches (e.g., PB). Buying over shorter planning horizons increases inventory turns with VF averaging at 18.3 vs. 5.0 for the same ASINs in AFN (September 2019). With OOS situations, VF's low VLT minimizes impact to CX (at the tradeoff of ship cost) while driving incremental revenue and maintaining instock levels. VF has proven its low VLT benefits through the following examples. **Peak 2018:** Between weeks 50-51, customer demand for 23 Mattel and Fischer Price (M&FP) deal ASINs increased by 35% vs. plan, depleting inventory levels across AFN. With an 18.0 day VLT in AFN, three VF M&FP nodes (CA, TX and PA) were utilized for rapid replenishment with a VLT of 0.75 days. The ability to provide national coverage through these nodes and maintain instock availability for all 23 ASINs generated an incremental topline revenue of $0.2MM. For Peak 2019, VF

[2] 41% of inbound volume has an average VLT of 1.68 days, 33% at 4.07, 19% at 2.78, and 7% of volume from 1 vendor (Phillips) at 8.87 days.

will have seven[3] Toy nodes with a target VLT of 1.0 day vs. projected AFN VLT of 17.0 days. **March 2019:** For the launch of PB Belei Wipes, actual demand was 800 units vs. a weekly forecast of 200. VF was utilized starting the second day of launch as inventory was depleting across AFN. VF fulfilled 100% of demand with a 0.25 day VLT (vs. 12.8 VLT for AFN) for the remaining four-weeks. VF maintained a RepOOS rate of 0.01% and a 98.2% FT instock rate[4] (+1,110 bps vs. new item averages, +320 bps vs. 2019 OP1 goal). **Prime Week 2019:** Due to an average VLT of 7.7 days for Reckitt Benckiser (RB), two last minute deal-of-the-day (DOTD) ASINs were inbounded into VF with a 0.25 day VLT, increasing total prime eligible inventory by 25%. During Prime Week, demand for these ASINs increased by a weekly average of 9.1K and 6.3K units vs. the trailing six-week average, with VF fulfilling 27% and 35% of the overall demand for the respective ASINs. The ability to quickly raise inventory levels through VF for these two ASINs, contributed to an increase in instock percent by +8bps and +23bps and FT instock rates +825bps and +25bps. **Prime Week 2019:** Demand projections for two Procter & Gamble (P&G) ASINs increased by 20% leading up to Prime Week with an AFN VLT of 9.5 days. Although these ASINs were not part of VF whitelist[5], seven P&G VF nodes were leveraged to inbound 6K units. During Prime Week, sales for these two ASINs increased by +14.5% and +9.4% vs. the trailing six-week average, with VF handling 25% of the national demand during a three day promotional period. **August 2019:** At the end of July 2019, eleven Pepsi Pure Leaf high velocity ASINs were facing OOS issues across AFN, as improper catalog dimensions resulted in vendor chargebacks. Pepsi and Retail have removed these ASINs from AFN and VF is being leveraged to maintain instock availability and prevent OOS until proper catalog dimensions are reflected (ETC - October 2019). Since week 31, VF has shipped 6.1K units weekly, while average instock percent and FT instock rate have increased by 16% and 36% vs. the trailing six-week averages.

Subscribe and Save (SnS) – Provide a Low Cost Fulfillment Solution for SnS: SnS's vision[6] is to enable global customers to replenish everyday essential items at any frequency. In 2019, SnS is projecting an OP Margin of 5.3% (+3.4% YoY) and an OPPU of $1.04 (+160% YoY), across 178M units (+4% YoY). In 2020, this is projected to grow to 5.5% and $1.11, across 216M units (+21% YoY). Of the 178M US units shipped in 2019, VF-eligible ASINs across the AFN and VF network will comprise 56.33M (31.6%) of customer shipments (40.0M AFN and 16.3M VF). While there is CX benefit to SnS, it creates a unique fulfillment challenge, where to protect the SnS experience, orders are encumbered to FCs and held in inventory up to nine days before shipment; known as shipment generation no earlier than (SGNE). This ensures that inventory is secured for our customers while constraining earliest FC ship date (SGNE) to safeguard against a bad CX as a result of shipping early. By leveraging free space within the VF network and utilizing VF's fast replenishment capabilities, overall SnS inventory position in AFN can be reduced, specifically for high SnS penetration ASINs (Table 3). This will enable capacity in AFN for a wider selection of inventory in support of 1DC. VF will partner with SCOT and SnS teams to determine tradeoffs for intentionally fulfilling more SnS orders through VF.

Table 3 - SnS AFN vs. VF Cube with Turns

			ACU		Inv Cube		Inv Units		Shipped Units (4 WK AVG)			Inv Turns	
	SNS %	ASINs	VF	AFN	VF	AFN	VF	AFN	VF	AFN	Total	VF	AFN
Non-Sort	20%	343	1.62	1.18	43,442	485,456	26,763	410,496	15,209	53,957	69,166	29.6	6.8
	30%	161	2.09	1.56	95,407	1,464,184	45,643	938,050	34,134	109,743	143,877	38.9	6.1
	40%	59	1.30	1.36	131,334	1,025,962	101,019	755,245	50,718	81,249	131,967	26.1	5.6
	50%+	39	2.64	2.46	102,607	1,231,208	38,913	501,182	22,876	44,577	67,453	30.6	4.6
	Total	602	1.76	1.61	372,790	4,206,809	212,338	2,604,973	122,937	289,525	412,462	30.1	5.8
Sort	20%	82	0.12	0.16	21,770	598,452	181,668	3,648,821	72,552	455,584	528,136	20.8	6.5
	30%	109	0.07	0.15	8,404	361,557	124,543	2,392,832	41,453	249,185	290,638	17.3	5.4
	40%	62	0.21	0.22	9,928	303,546	47,695	1,355,183	31,632	113,175	144,806	34.5	4.3
	50%+	25	0.10	0.16	2,706	76,761	26,381	493,192	19,423	36,663	56,086	38.3	3.9
	Total	278	0.11	0.17	42,807	1,340,315	380,287	7,890,028	165,060	854,606	1,019,666	22.6	5.6

*Inventory Snapshot of 9/05 with trailing four weeks of shipped units data

Section 4: Outbound Transportation

In In January 2019, VF utilized 3P carriers (e.g., FedEx, UPS, USPS, and Ontrac) for 86% of units and IC (e.g., Sort Center – AMZL, USPS last mile) for the remaining 14%. As IC, connectivity has expanded and due to economies of scale, VF has shifted from a 14% utilization, to an August 2019 average utilization of 41%. Actions taken to improve IC utilization included launching previously unconnected nodes to SC and connecting to SC hubs to increase zip code coverage. Currently, 95% of VF nodes are connected to SC (+90% YoY), with plans to reach 100% by Peak 2019. All future VF launches will be connected to SCs at launch. August YTD, VF has maintained an average Delivery Estimate Accuracy (DEA) of 97.5% (vs. 96.8% for AFN), with an OB TransCPU average of $4.36

[3] 3 M&FP, 2 Hasbro, 1 SpinMaster, and 1 Melissa and Doug.
[4] FT instock rate represents the percentage of total GV that had a FT message where the ASIN was instock.
[5] Whitelist represents the ASINs a VF node is fulfilling, based on what the vendor has available within their distribution center.
[6] SnS vision and 2019, 2020 OP Margin, OPPU and shipped units is derived from SnS committed OP1 plan.

(+1.8% vs. January 2019). For the same ASINs, AFN utilized IC 80% (including USPS final mile), with an OB TransCPU average of $3.79 (-8.0% vs. January 2019). Table 4 includes main drivers for the OB TransCPU delta between VF and AFN. Drivers include: i) difference in miles shipped (VF averages 432 miles vs. 125 in AFN) as a result of VF shipping longer distances due to limited topology and fulfillment benefits within F2P; ii) VF has an average ship ACU of 0.92 vs. 0.56 for AFN, as a result of VF shipping larger ACU products (e.g., Diapers, Paper Towels, Toilet Paper); and iii) deprecation of FedEx (week 27) increased costs, relative to AFN. FedEx accounted for 15% of all VF units prior to deprecation with an ACU of 2.11. Volume that shifted to UPS (65% of FedEx volume) was subject to large package surcharges, additional handling fees and higher base rates, which increased costs by an average of $1.25/u. VF will expand utilization of IC to maintain parity with AFN-wide targets 75% by 2022 (Amazon managed last mile).

Table 4 - August 2019 Ship Option Comparison (same ASINS)

Ship Option	VF	VF OB TransCPU	Avg. Miles	VF ACU	AFN	AFN OB TransCPU	Avg. Miles	AFN ACU
SnS	53%	$ 4.83	500	1.14	12%	$4.37	108	0.69
Second	19%	$ 3.53	339	0.42	35%	$3.40	128	0.37
1DC	14%	$ 3.85	254	0.54	37%	$3.41	79	0.34
Standard	8%	$ 5.35	491	1.68	14%	$5.24	250	1.46
Super Saver	8%	$ 3.03	461	0.54	1%	$3.45	120	0.35
Total	**100%**	**$ 4.36**	**432**	**0.92**	***98%**	**$3.79**	**125**	**0.56**

*2% delta for AFN includes International and Same Day Shipments

Section 5: Operations

Operating and Labor Structure: A single shift structure is operated 7-days a week for 65% (92% by Peak 2019) of VF nodes. To increase flexibility in handling demand fluctuations and optimize labor usage, the VF network will maintain a ratio of 75%-to-25% (+15% in temp utilization YoY) Blue Badge (BB) to Temp Associates and will to move to a 70%-to-30% ratio by 2020. At least two BB Associates are hired at every VF node and rotate between process paths (e.g., inbound, stow, pick, pack, and ship) throughout the day, which increases labor efficiencies. Any additional headcount needed is filled by giving the first preference to hiring Temp Associates then transitioning to BB as throughput stabilizes. More than two BB Associates are hired in areas where Temp labor is difficult to acquire (e.g., TX and MO) and in instances where nodes are working 7-days of the week. During Peak, only Temp Associates are hired to support volume surges. 75% of VF nodes are clustered together (<35 miles apart), which increases the VF network's ability to labor share Associates between nodes, similar to how an FC would share between process paths. The ability to utilize Associates across many process paths while streamlining operational processes and hiring Temp labor with labor share, is a main driver to achieving a VCPU benefit of $0.65 compared to AFN.

Section 6: Next Steps

VF is positioned as a fulfillment solution that compliments AFN by alleviating capacity constraints in the Non-Sort network, mitigating OOS risks, fulfilling selection that is non-speed sensitive (e.g., SnS), and absorbing the complexity of managing growing businesses such as PB. The following opportunities have been identified to drive incremental value and further our impact.

Project Salmon – Exposing Vendor Availability Feeds to Improve Buying and Placement Decisions. Project Salmon will improve buying through the introduction of inventory availability signals from the vendor. This ability will enable SCOT systems to plan with supply availability and location awareness resulting in higher confirmation rates and reduced VLT. In the current state, SCOT systems do not have visibility into a vendor's available inventory or locations and assumes an unlimited supply when making buying decisions. The lack of visibility causes issues across our network. Within VF specifically, confirmation rates have averaged 72% (2019 YTD) resulting in sub-optimal utilization of vendor-provided free space with fullness averaging 70%. With Salmon, our goal is to achieve a confirmation rate of +90%, reduce program VLT to under 1.0 day (2.98 days 2019 YTD), and improve fullness to 95%. A Minimum Viable Product (MVP) solution was developed in Q4 2018 in partnership with AFT – FaaST and SCOT SORES/Workflow teams as proof of principle for Project Salmon. This solution was deployed across 29 nodes and four Vendors (P&G, Kimberly Clark (KC), Georgia Pacific, and FQ) and provided a daily replenishment signal with manual inputs that simulated the location specific to available ASINs. August YTD FulCPU savings of $4.32 for the MVP selection of 214 Salmon ASINs[7] has been achieved across KC, P&G and FQ with an 87.3% confirmation rate and 1.56 day VLT (see FAQ 1 for Vendor Engagement). The MVP solution will eventually be phased out as a scalable, automated approach that is developed in partnership with SCOT and AFT teams for 2020 deployment. In the long term, Salmon's upstream visibility will optimize buying and placement into both VF and AFN. Currently, the SCOT Supplier Operations and PB teams are acquiring upstream information similar to what Project Salmon will provide in order to: i) provide vendors accurate ship windows that take into account product locations; and ii) gain visibility into future

[7] Georgia Pacific excluded due to vendor supply chain issues impacting confirmation rates across AFN and VF.

production and availability for PB ASINs. We will share learnings from the automated Salmon solution with SCOT and PB for application within AFN.

Private Brands – Reducing Costs and Providing a Platform for Growth. In 2019, domestic Consumables PB (CPB)[8] has a GMS target of $325MM[9], growing to $835MM in 2020. PB is an important but complex business that faces headwinds in terms of ability to quickly scale new partners and products, increased VLT (VF VLT of 1.98 days and AFN of 10.25 days for same vendors) creating long replenishment cycles, increased supply chain costs (primarily fixed), and inaccurate forecasting due to volatile demand. To help minimize the impact of these headwinds, VF began collaborating with PB by launching two nodes (FQ – *Presto Paper Towel* and Nice Pak – *Mama Bear Wipes*) in Q3 2018. By the end of 2019, VF will have expanded partnership to six PB-specific and fourteen joint PB and core Consumables nodes. The VF network's ability to provide supply chain cost savings for PB, is evident through an August YTD FulCPU savings of $5.01 (PB ASINs). Fixed savings ($4.27/u) provides the largest benefit as a result of VF PB turns at 24.1 vs. AFN PB turns at 4.7 (Table 5). IB savings ($1.01/u) and variable savings ($1.04/u) are gained from eliminating Amazon freight costs (WePay vendors) and operating with a vendor-funded labor model (no labor cost to Amazon). To drive savings in variable costs, vendor-funded labor will be a prerequisite for future PB node launches. The tradeoff for PB is OB TransCPU (including UPB impact), where VF PB ASINs lose -$1.32/u vs. the same ASINs in AFN. Drivers to increased OB TransCPU are: i) VF shipping PB SnS orders over long distances (582 average miles for VF vs. 116 for AFN); and ii) VF ship ACU of 1.86 vs. 1.13 for AFN.

Table 5 - AFN vs. VF Cube and Turns for PB ASINs

		ACU		Inv Cube		Inv Units		Shipped Units (4 WK AVG)			Inv Turns	
Sort Type	ASINs	VF	AFN	VF	AFN	VF	AFN	VF	AFN	Total	VF	AFN
Non-Sort	44	1.98	1.66	135,120	1,061,016	68,177	640,823	32,648	66,819	99,467	24.9	5.4
Sort	11	0.35	0.44	3,123	166,670	8,934	377,470	3,051	25,793	28,844	17.8	3.6
Total	55	1.79	1.21	138,243	1,227,687	77,111	1,018,293	35,699	92,612	128,310	24.1	4.7

*Inventory snapshot of 9/05 with trailing four weeks of shipped units data

Two examples that underline the VF network's ability to decrease complexity for PB are listed below.

Product Launch Impact. FT for PB ASINs (55) enrolled[10] in VF has been 96.2% trailing twelve months (+340 bps vs. non-VF ASINs and +162bps vs. 2019 OP1 goal). Demand ramp-up for new selection is a primary driver to increasing YoY PB topline growth. This demand penetration is non-linear due to time spent building discoverability and search relevancy. To accelerate sales, PB utilizes "cold-start" couponing to drive a path to search rank maturity. Due to highly variable SCOT forecasts for new ASINs, couponing in the initial weeks can drive OOS challenges; new item FT rate at 87.1%, -570bps vs. total CPB. CPB launched the Belei brand in Q3 2019 and were able to achieve a FT% of 98.2 leveraging VF (+1,110 bps vs. new item averages, +320 bps vs. 2019 OP1 goal).

Treehouse Launch (September 2019). Treehouse is a top five CPB supplier (2019 YTD GMS 8% of total CPB) whose primary subcategory is coffee pods. In early 2019, Victor Allen coffee pods (national brand owned by Trilliant) started aggressive pricing, promotions and Amazon Marketing Services (AMS) activities to gain OPP share on Amazon. As a result, Victor Allen attained 2.6% share in Q1 (+65 bps QoQ), while Solimo's (PB) coffee pods share reduced to 4.1% (-10 bps QoQ). Based on this and PB's long-term volume aspirations in the coffee pods space, PB decided to price Solimo coffee pods at a -3% discount vs. Victor Allen, resulting in an average sale price decline of -$4.80. Dropping prices to remain competitive has led to a contribution profit less fixed cost per unit (CPLF/u) of $1.30 and accretiveness of -611 bps (+601 bps excluding Display Ads, -99 bps vs. goal). With Solimo coffee pods moving to VF in October 2019, PB expects to improve their CPLF/u to $1.83 and accretiveness to -451 bps (+784 bps excluding Display Ads, +84 bps vs. goal).

[8] Less accelerator brands and Whole Foods Market.
[9] Five vendors account for 45.2% of total GMS. VF has nodes with all of these vendors and accounts for 22% of national demand.
[10] Expanding to 107 by October 2019.

FAQs

FAQ 1 - Why are vendors engaged with VF and initiatives like Project Salmon? VF continues to gain buy in and support from top-tier Retail vendors as a result of i) reducing transportation costs for vendors, specifically TheyPay vendors (large Non-sort vendors like P&G, KC and GP – who average an ACU of 2.0); ii) reduces variable and material costs associated to prep time for outbound shipments to Amazon; iii) enables vendors to improve key Retail metrics (FT conversion, RepOOS, FT GV, etc.) by utilizing VF replenishment capabilities, especially in OOS instances iv) improves vendors aggregate VLT, monitored by VOLT (vendors use improved metrics vs. peers for negotiations); v) enables joint savings and improvement to a vendor's P&L within Retail Categories; vi) ability to leverage VF to launch new products; and vii) to quickly identify and resolve ASIN specific issues (i.e. case pack quantities, bar code issues, etc.). Project Salmon provides further benefit by tackling one of the biggest headwinds vendors face today – purchase order variability. Salmon solves order variability by providing Amazon visibility into the vendors' selection availability by warehouse. This allows vendors to better plan and stock inventory in support of VF vs. how they operate in a reactive manner today. VF drives savings for both vendors and Amazon, these joint savings, specifically for vendors, enable them to improve profitability and their overall engagement with Amazon.

FAQ 2 - What are additional benefits Amazon Retail gains from VF? Non FulCPU Retail gains through VF include i) rapid replenishment abilities through reduced VLT, reducing total inventory levels for vendors that VF have high penetration with (i.e. FQ), minimizing OOS risk and providing a mechanism for Retail to launch new products (i.e. Private Brands Belei wipes); ii) enhancing vendor relationships, specifically Private Brands, as VF is a joint venture to reduce overall costs for both the vendor; and Amazon iii) providing incremental ContraCogs credits when throughput targets are achieved, specifically for KC. KC provides Amazon Retail $0.25/u credit when throughput thresholds are achieved and iv) ability to utilize VF to pilot supply chain initiatives such as direct mill[11] and advertising initiatives (i.e. VF puts Toy advertisements from Retail and the vendor in boxes) for Prime and Peak.

FAQ 3 - What are the most hotly debated topics? Two hotly debated topics include i) how should F2P model VF's fixed cost avoidance; and ii) what are the main differences between VF and Direct Fulfillment (DF). Fixed cost avoidance is the leading financial value proposition for VF and today is not modeled in our placement and fulfillment systems (F2P). This is because fixed cost is considered sunk costs for FCs, as outside of Prime and Peak FCs run underutilized. However, as AFN continues to face capacity constraints with increased demand, modeling this benefit to allow us to fully utilize VF capacity is essential. Modeling fixed cost avoidance in F2P for VF is an ongoing discussion between Finance teams, VF and F2P tech but given the complexities of this undertaking, no definitive plan has been determined. For VF and DF differences, VF operates 7-days a week with a 2HR FT SLA and enabled for 1DC, while utilizing IC (42%). VF is primarily focused on fulfilling high velocity head and body selection (mainly Consumables – 65%) that overlaps with AFN. DF operates 5 days a week with a 6HR FT SLA and has only 2% of nodes enabled for 1DC, while utilizing only AMXL as an IC. DF historically has been focused on unique selection that does not overlap with AFN. However, more recently, DF has expanded into head and body selection across top vendors (KC) to understand their capabilities. Today 950 ASINs overlap between VF and DF, with Media ASINs accounting for 775, Toys with 115 and Consumables with 60. Of the 950 ASINs, VF fulfills 15.4% (-23% since June) of national demand as of August-19, with DF fulfilling for 6.6% (+120% since June). Finally, VF has enhanced control over CX, through utilizing Amazon internal warehouse management tools and Amazon labor. While we believe there are fundamental differences and value propositions for each program, the increased overlap and change in DF strategy is driving confusion across SCOT, Retail and with the vendors themselves. By Q4-2019 we will partner with DF and align on clear go to market strategy that eliminates redundancies.

FAQ 4 - What does F2P's OB TransCPU simulation tell us about costs if VF did not exist? F2P simulations indicate that the OB TransCPU delta between VF and AFN is $1.16 (+0.64 larger than current reporting – including UPB). The result is based on a F2P online simulation where VF supply does not exist. This allows us to compare the OB TransCPU of a customer order assigned to a VF node in production against the cost had the same order been assigned to an AFN node instead. The simulation provides a more accurate model of VF's impact on consolidation, distance, and carrier mix. Previously, F2P simulation data was not a reliable source for VF outbound reporting as simulations are run against FTAs (the first plan after an order is placed) instead of LTAs (the last plan before an item is shipped). FTA data can be misleading due to changes in the FC assignments between the first and last plan. This is especially prevalent for SnS orders, which have a longer planning lead-time. However, in the updated analysis F2P bridged the outbound rates between FTA and LTA to account for this limitation. We are working with SCOT to better understand the gap and integrate the calculation into a streamlined reporting process.

FAQ 5 - Do core SCOT systems recognize and plan for VF nodes? Yes, today our core SCOT systems recognize and plan placement, buying, and fulfillment out of VF nodes. However, we have identified opportunities across SCOT to ensure VF benefits are adequately represented. We are partnering with Profit Intelligence to ensure that our upstream systems, such as ContraCogs, have

[11] Direct mill is a concept of shipping directly from a production plant into a VF cage – bypassing staging in a vendor's DC.

the ability to delineate and capture benefits from VF nodes vs. FCs. This will feed downstream system planning systems the correct cost savings by program. With Continuous Buying, we are partnering to deep-dive buying logic to understand what levers we have for our systems to capture VF VLT benefits. From a Placement standpoint, we led an initiative to update the legacy VF planning algorithm, focused on maximizing the available free space within VF nodes. With F2P, we are partnering on several initiatives i) fixed cost benefit recognition for order assignment; and ii) right sizing FC variable costs within F2.

FAQ 6 - How are we thinking about launching Toys nodes going forward? We have traditionally launched year round Toys nodes but in 2020, we will bias towards launching Toys nodes only for Peak, unless a Toys vendor has proven year round demand and stable selection (e.g., Legos). Given our short launch lead-time (average of seven weeks), we have the ability to quickly launch Toys nodes for Peak, where we are able to generate the highest ROI and then quickly shut down during Q1. Initial conversations with both Retail and vendors show equal willingness in this initiative, as off Peak demand volatility headwinds are jointly faced.

FAQ 7 - Are we at risk to giving away intellectual property (IP) within VF? No, we are not at risk to giving away IP and have implemented processes to ensure we are adequately protected. There are two IP considerations we manage i) risk to software and ii) risk to physical operations. For #i, we have implemented access controlled workstations within VF nodes, managed by Amazon approved associates and only accessible through Amazon standalone Wifi and VPN. For #ii, we have several measures in place to protect both our operations and equipment within the node. We deploy blackout netting across VF nodes in instances we are concerned about processing capabilities or selection within the VF nodes. We lock all VF nodes outside of working hours, with access given to select vendor points of contact for emergency purposes. Processing materials will be locked in storage cabinets, while laptops will be taken home by the PA. We utilize cost circuit televisions to monitor activity in and outside of our nodes (average of 100-day retention policy). In addition to these measures, we have loss prevention, safety and IT audits performed quarterly across nodes.

FAQ 8 - How do we leverage / knowledge share with other Geographies? We partner closely with other geographies on best practices, initiatives, and operational standards. This collaboration occurs in many forums, from monthly syncs to our annual summit event. We have worked closely with the EU to help them replicate our Project Salmon roll out, Canada on how to best manage and optimize for SnS, India on how to best leverage the software capabilities of the warehouse management system, and with Japan on how they are optimizing their transportation strategy. For new country launches, such as Australia, Brazil, UAE and Mexico, we have closely partnered with these teams to share lessons learned, help them identifying the best build plan for their countries, and aided them in scoping which variables will yield the largest ROI, including country specific nuances (topology, capacity and product mix).

285 ## Appendix A

286 VF Selection Criteria

VF Vendor and ASIN Selection Criteria		
Type	Criteria	Description
Vendor	7 Day Operations	To increase our 1DC penetration, we will only launch with vendors who allow us to operate 7 days a week.
Vendor	Evergreen Selection	We will prioritize partnering with vendors that have evergreen selection (i.e. Consumables), as they provide consistent demand year round (better labor planning). For Peaky vendors (Toys), we will evaluate the financial opportunity to launch a node for only Peak.
Vendor	Funded Labor	To continue to drive incremental savings, we will partner with vendors who offer to pay for our labor. This will allow us to stay competitive as costs are driven down across AFN.
Vendor	Partner with Tier 1 or 2 Vendors	Tier 1 vendors are vendors that ship >4M units annually and tier 2 vendors are ones that ship between >2M and 4M units annually. Targeting these vendors will ensure that we are biasing towards high velocity vendors to maximize turns through VF.
Vendor	Ability to Open Multiple Nodes	We will only partner with vendors who have the ability to open >2 nodes, as it increases our topology presence allowing us to not only keep transportation costs low but increases our 1DC penetration.
Vendor	Location	We will evaluate vendors based on location and favor vendors located in major customer demand markets to optimize for ships costs and proximity to customers.
Vendor	10% YoY Growth and CP Positive	We will partner with vendors who show continued YoY profitable growth. This will ensure that our nodes are not impacted due to vendor negotiations and that we can maximize the space within VF year round.
Vendor	Multiple Product Brands	We will bias towards vendors who span multiple Categories and GLs (i.e. Nestle), to leverage unit and savings entitlement. Not only is this beneficial for VF but increases economies of scale and reduces costs for Retail GLs within the node.
Vendor	Private Brands	We will prioritize partnering with PB, providing a mechanism to reduce supply chain complexity and costs.
ASIN	Subscribe and Save	We will look to partner with vendors who have ASINs that are high in SnS penetration, as this product is expensive to hold across AFN. We will leverage our free space to drive down costs for SnS items.
ASIN	Non-sort SIOC Preferred	Where possible, we will focus on selection that is Non-sort in nature, as it provides the greatest savings and is the most efficient to process operationally. Focusing on SIOC selection will also reduce our ability to create split shipments.
ASIN	Cube	For sort selection, we will bias towards higher cube sortable selection ranging from 0.25 - 1.1 and for non-sort selection, bias towards higher cube product 1.1 - 4.0.
ASIN	Affinity to ship with similar ASINs	We will look to partner with vendors who have products that ship together (i.e. P&G Pampers and Tide), as it will decrease our UPB impact compared to AFN.
ASIN	Operational Requirements	We will evaluate each ASIN to make sure it meets our operational constraints (i.e. mechanical equipment, weight restrictions, Hazmat status, and ASINs parent/child relationships) to ensure that we can fulfill selection in the most efficient way.

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288 ## Appendix B

289 Number of nodes by Category – A single node may have many vendors code across many Categories

Category	2017	2018	2019
Consumables	23	28	32
Hardlines	11	11	12
Consumables / Private Brands	0	7	7
Consumables / Private Brands / Hardlines	6	6	6
Private Brands	0	2	6
Consumables/Hardlines	3	3	3
Total Nodes Year End	43	*57	**66

**6 nodes shut down throughout 2018*
**5 nodes shut down throughout 2019*

290 Number of VF Vendors

Number of Unique VF Vendors	2017	2018	2019
Total	12	24	28

291 ## Appendix C

292 Operations Headcount

Table 6 - Operations Headcount			
Breakout	East	Central	West
Number of VF Nodes	26	20	22
Senior ROM	1		
ROMs	1	1	1
L4-L5 AM	6	5	6
Blue Badge Associates	195		
Temp Associates	75		
Total Headcount	*291		

*Current headcount projected to increase to 320 for Peak

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